HAYATO OTSUKA

SENIOR FULL STACK ENGINEER | PYTHON, DJANGO, JAVA, REACT

hayatootsuka.dev@gmail.com | Tokyo, Japan

PROFESSIONAL SUMMARY

Senior AI Full-Stack Engineer 12+ years of experience designing, implementing, and leading distributed enterprise technology, high scale microservices (Java, Python) and modern reactive web stacks (TypeScript, React), with deep expertise in the engineering & operationalization of generative AI, LLM orchestration & execution pipelines, and vector-backed search pipelines embedded into production full-stack flows. Growth-oriented engineering leader, able to scale teams and reduce cloud costs by 20%+, with global enterprise experience.

TECHNICAL SKILLS

- **Languages:** Python (FastAPI, PyTorch, LangChain), Java (Spring Boot 3, JVM tuning), TypeScript, JavaScript, SQL
- **Frontend & UI:** React, TypeScript, Redux, Tailwind CSS, AI-Native UI/UX Patterns
- **AI & LLM Stack:** LangChain, LlamaIndex, OpenAI/Anthropic APIs, RAG Architectures, Vector Databases (Pinecone, pgvector, Milvus), Local LLMs (Ollama, vLLM)
- **Security & Governance:** OWASP Top 10, OAuth2/OIDC, LLM Guardrails (LlamaGuard, NeMo Guardrails), Semantic Caching Security, RBAC/ABAC, Dependency Scanning
- **Testing & Quality Assurance:** PyTest, JUnit, Playwright, Cypress, CI/CD Test Automation, RAG Evaluation (Ragas, TruLens), Mocking/Stubbing Frameworks
- **Cloud & DevOps:** AWS (ECS, EKS, Lambda, S3), Azure, Docker, Kubernetes, Terraform, GitHub Actions CI/CD
- **Architecture & Data:** Distributed Systems, Event-Driven Architecture (Kafka), PostgreSQL, MongoDB, Redis, GraphQL, REST APIs

PROFESSIONAL EXPERIENCE

Senior Software Engineer | *Timee, Inc*

May 2024 – Oct 2026

- **AI-First Platform Architecture:** Developed and implemented a company-wide RAG and agentic workflow platform with Python/ FastAPI/ Lang Chain, reaching more than 45,000 internal users.
- **Full-Stack Modernization:** Rebuilt several legacy portals for customers using TypeScript, React, and Next.js, which brought the load times of the pages from 3.8s (average) down to 1.1s.
- **Microservices & Java/Python Interop:** Led the porting of core monolithic transaction engines into distributed Java (Spring Boot) and Python microservices on AWS EKS, which delivered a 3.5x throughput gain.
- **Cost & Performance Optimization:** Applied vector caching and query optimization layers in order to lower the cost of LLM token inference and cloud infrastructure cost.
- **Security & Guardrails:** Created set of production ready semantic caching pipelines and input/output guardrails with Llama Guard to prevent prompt injection attacks and data exfiltration on all client application entry points.
- **Streamed Data Infrastructure:** Integrated SSE and WebSocket's with Spring Boot and FastAPI for low latency tokens and intermediate agent thoughts from intermediate / signal layers to the UI layer.
- **Technical Governance:** Designed enterprise blueprints for multi-agent systems and a set of consistent AI-native UX components patterns for global engineering hubs.
- **Leadership & Strategy:** Mentored 3 junior developers on cross-functional product teams to adopt prompt engineering pipelines, CI/CD automated tests, and code quality standards.

Presales Engineer | *IBM*

Oct 2022 – Apr 2024

- **Real-Time Data & Backend:** Created and managed Java and Python high concurrency REST APIs for Java and Python containing over REST APIs with 4.5M+ daily transactional events.
- **Frontend Overhaul:** Architected interactive TypeScript React dashboards with real-time Web Socket telemetry monitoring, increasing user engagement per-user ratio.
- **Machine Learning Integration:** Integrated scikit-learn and PyTorch micro-models in backend data pipelines for automatic anomaly detection, decreasing false-positive alerts.
- **CI/CD Automation:** Developed automated testing gates in GitHub Actions and dockerized local environments, zero rollbacks to deployment.
- **Infrastructure as Code:** Developed Terraform modules to provision AWS resources (RDS PostgreSQL, ECS cluster, S3 storage), resulting in 60% reduction of the time-to-market for AWS infrastructure.

- **Event-Driven Architectures:** Re-architected data ingestion tier using Apache Kafka to disengage analytical (high volume) traffic from primary relational storage architecture.
- **Team Scaling:** Led the scaling of the distributed engineering organization through seamlessly onboarding & training incoming engineers and owning the entire technical feature scoping from inception to launch.

Senior Software Engineer | *Mercari, Inc* Jun 2018 – Sep 2022

- **Core Development:** Developed and delivered full stack web features using Java (Spring) and Python (Flask) and Angular/TypeScript.
- **Database Tuning:** Migrated legacy schemas from Microsoft SQL Server to PostgreSQL, leading to increase in performance for complex analytical queries.
- **API Standardization:** Wrote company-wide OpenAPI/Swagger specs for other teams to build against for customer-facing software products to decouple development teams.
- **Performance Monitoring:** Scripted Datadog dashboards and APM alerting logic, which cut the Mean Time to Resolution (MTTR) in half for production incidents of significance.
- **State Management:** Updated fast modular web applications with Redux and NgRx to handle substantial client-side caching.

Software Engineer | *Money Forward, Inc* Feb 2017 – May 2018

- **Backend Systems:** Built backend services in Java (Spring MVC) and Python, included scripts for processing for large data sets.
- **Frontend Migration:** Moved old legacy jQuery frontends to TypeScript & early stage React components with a better and more reactive user experience.
- **Testing & Quality:** Developed 90%+ code coverage through the establishment of thorough Junit/Pytest automation suites across key business lines.
- **Data Integration:** Integrated several third-party partner data feeds into core corporate storage layers using a custom-built Python and SQL ETL toolset.
- **Collaboration:** Worked with UX/UI design team to develop library of reusable components that reduced front-end layout engineering effort by 30%.

Software Engineer | *Sansan Corporation* May 2014 – Jan 2017

- **Full Stack Development:** Created web portals using Django & vanilla JavaScript, Bootstrap, handling large relational schema in MySQL.
- **Testing & Quality:** Developed unit test suites using unit test, a built-in Python module, to prevent regressions from being passed on to staging systems.
- **Microservice Optimization:** Decomposed the payment gateway and billing microservices integrated with third-party payment gateways to support higher performance via error-handling and transaction validation.
- **Technical Documentation:** Maintained engineering documentation by giving archipelago maps to assisting junior team members on function.
- **Infrastructure Automation:** Automated backup plans and SSL/TLS terminations for all of our company's servers that are publicly accessible.

EDUCATION

Bachelor of Science in Information Technology – *Tokyo Denki University* Sep 2011 – May 2015